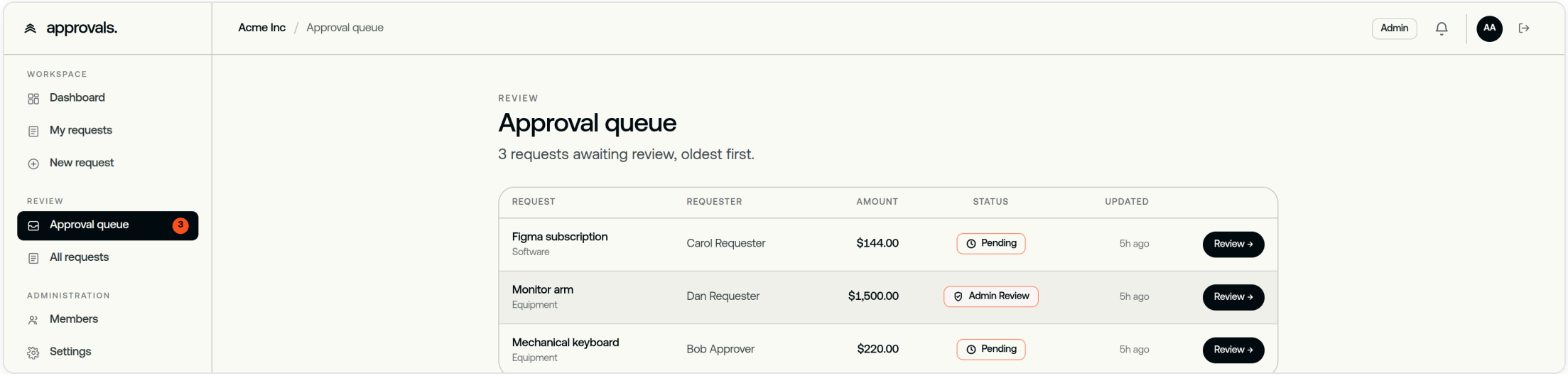
approvals.

Request, review, and decide on expenses — multi-tenant, secure by default, and fully responsive from desktop to phone.

72-hour take-home submission

NEXT.JS · SUPABASE (POSTGRES + RLS) · VERCEL · SENTRY



The problem

Small teams approve money over email and chat — no trail, no accountability, no status. In fintech, an approval with no immutable record is a compliance failure, not an inconvenience.

BEFORE — EMAIL & CHAT

expenses · Slack

Dana
can someone approve my $1,500 laptop? 🙏

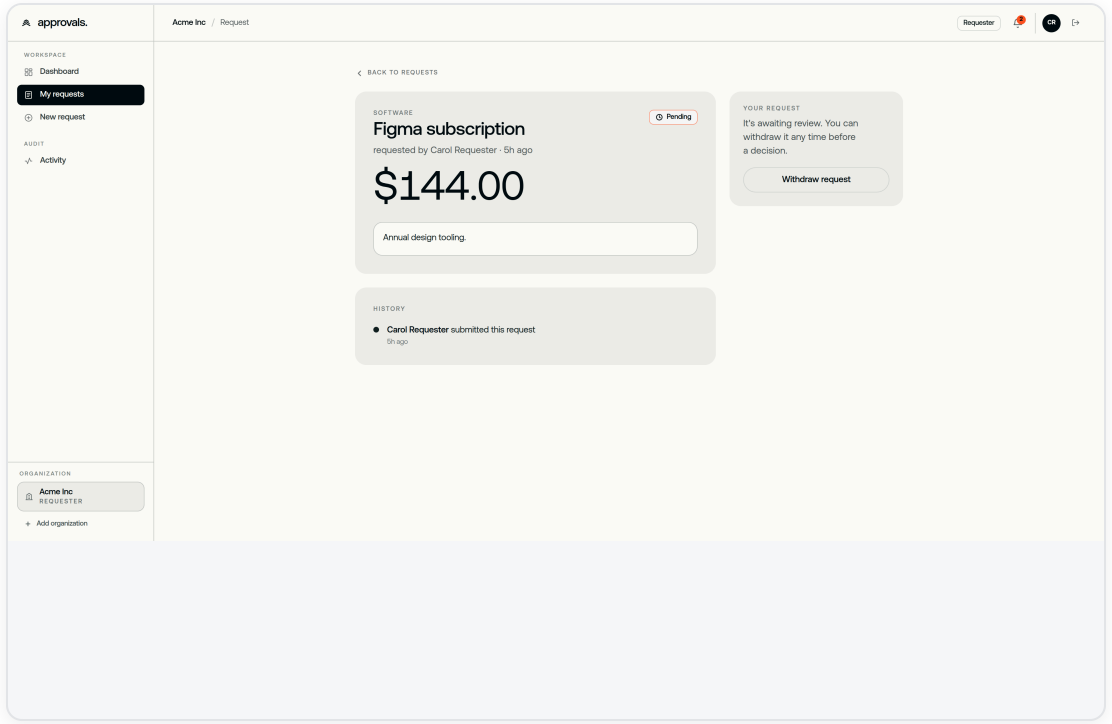
You
who approves these now that Sam left?

Priya
i think i ok'd it last week? or was that the monitor

Dana
did this get approved?? finance is asking 😬

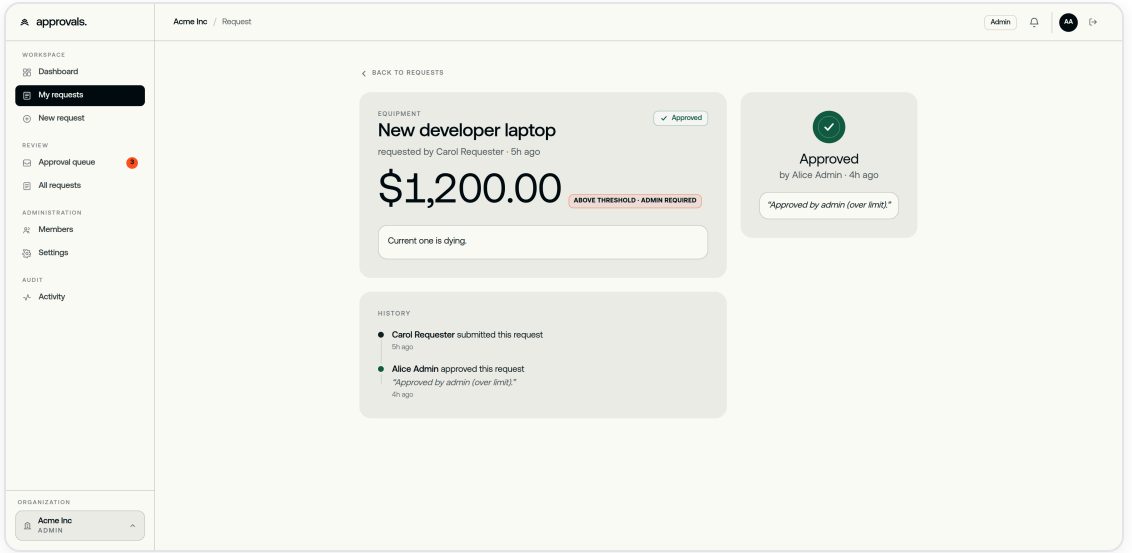
You
no idea — check the email thread

AFTER — ONE AUDITED RECORD



What I built

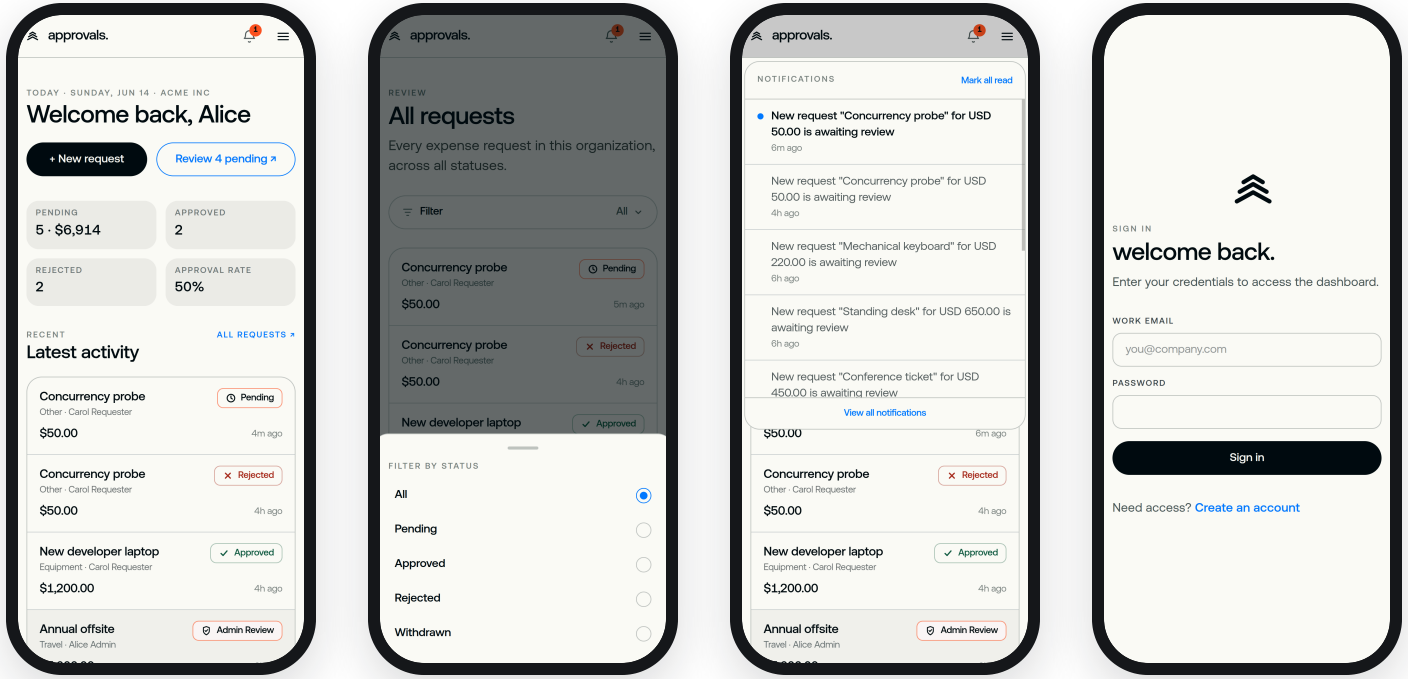
- Multi-tenant expense approval app: request → review → decision
- Three roles: **requester, approver, admin** (admins can also submit)
- Full lifecycle: **pending → approved / rejected / withdrawn**
- Resubmission creates a **new** request — the rejection stays on record permanently
- Immutable audit trail on every state change



REQUEST DETAIL — EVENT TIMELINE (ADMIN)

Designed for every screen

The same flat, ink-on-paper system reflows to any width — no feature dropped, no separate mobile theme, an approver can clear the queue from a phone.



DASHBOARD — FILTER — BOTTOM SHEET — NOTIFICATIONS DRAWER — SIGN IN

Reflow, not shrink

Tables become tap-friendly cards; the four-stat row stacks 2×2; nothing scrolls sideways.

Native-feeling patterns

Filters open as a bottom sheet with a grab handle; notifications as an anchored drawer — not a new page.

Touch targets & safe areas

44px minimum hit areas, `100dvh` sheets, and `safe-area-inset` padding for the home bar.

One system throughout

Same tokens, type and status language as desktop — accessibility and contrast hold at every width.

Data model

- Six core tables: **organizations · profiles · memberships · requests · request_events · invitations**
  (plus a notifications table for in-app alerts)
- **memberships** is the user ↔ org join — it carries the role
- Every domain table is scoped by **org_id** — tenant isolation at the data layer
- Money as **integer minor units (cents)** — never floats
- **request_events** is append-only — no UPDATE/DELETE policy exists

```sql
create table requests (
  id            uuid primary key,
  org_id        uuid not null references organizations,
  requester_id  uuid not null references auth.users,
  amount_minor  bigint not null check (amount_minor > 0), -- cents
  currency      text   not null,
  status        request_status default 'pending'
);

create table request_events (   -- append-only
  id          uuid primary key,
  org_id      uuid not null references organizations,
  request_id  uuid not null references requests,
  actor_id    uuid references auth.users,
  from_status request_status,
  to_status   request_status,
  note        text,
  created_at  timestamptz default now()
);
```

Excerpt — …_init.sql

Security architecture

- Auth via **@supabase/ssr httpOnly cookies** — JWT never in localStorage
- **RLS on every table**, default-deny
- Policies written **per operation** — SELECT, INSERT, UPDATE, DELETE separately
- A SELECT policy grants **no** write access — each verb explicitly allowed or denied
- Helpers **is_org_member / is_org_approver / is_org_admin** keep policies clean
- Service-role key is **server-only** — the browser holds the public anon key; RLS protects the data

```
-- SELECT: owner OR an approver in the org
create policy requests_select on requests
  for select to authenticated
  using ( requester_id = auth.uid()
          or is_org_approver(org_id) );

-- INSERT: only for yourself, only in your org
create policy requests_insert on requests
  for insert to authenticated
  with check ( requester_id = auth.uid()
               and is_org_member(org_id) );

-- No UPDATE / DELETE policy exists.
-- authenticated isn't even granted UPDATE —
-- status changes are impossible outside the RPC.
```

Excerpt — RLS policies on `requests`

The decision RPC

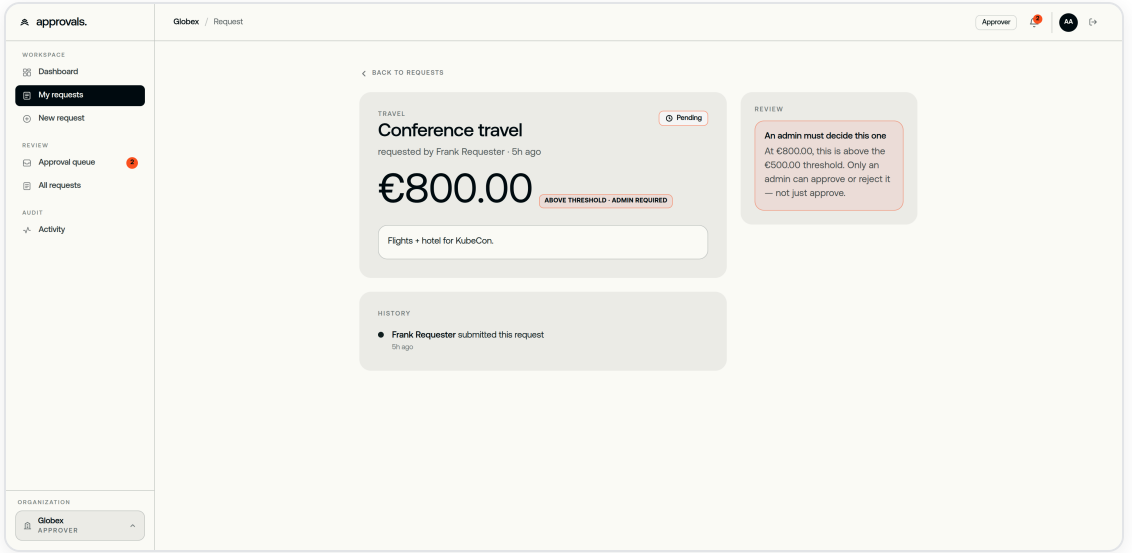
- **decide_request()** is a **SECURITY DEFINER** Postgres function
- One atomic transaction enforces: member with approver/admin role; **not the requester**; still **pending** (row locked); threshold routing; status flip **+** audit insert — both or neither
- Business logic lives in the **database**, not the API
- Can’t be bypassed via UI or raw API — **authenticated** isn’t granted UPDATE on requests

```
create function decide_request(p_request uuid,
       p_decision request_status, p_note text)
security definer set search_path = '' as $$
  select * into v_req from requests
    where id = p_request for update;        -- lock
  if v_req.status <> 'pending' then raise; -- 1 winner
  if v_req.requester_id = v_uid then          -- segregation
     raise 'you cannot decide your own request';
  if not is_org_approver(v_req.org_id) then raise;
  if v_req.amount_minor > v_threshold           -- threshold
     and not is_org_admin(org_id) then raise;
  update requests set status = p_decision …;    -- flip
  insert into request_events …;               -- + audit
$$;
```

Excerpt — `decide_request` (abridged)

Threshold routing

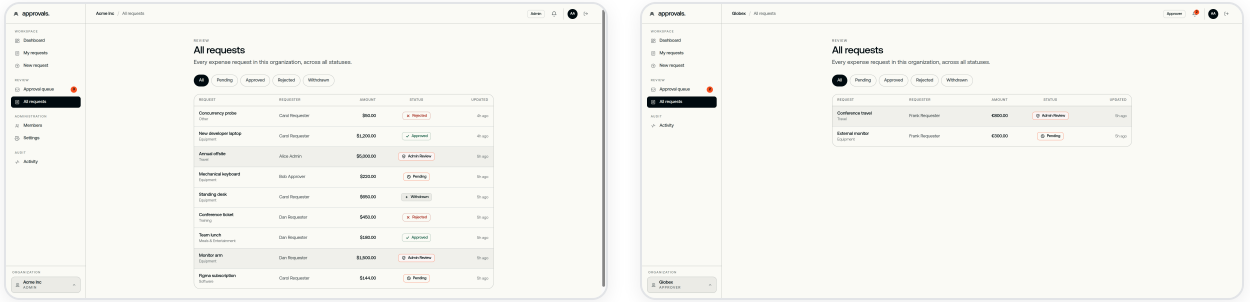
- Each org sets an **approval_threshold** (in minor units)
- **Under** threshold → any approver or admin may decide
- **Over** threshold → admin role required
- Enforced inside the RPC — the UI flag is informational only
- **Single-admin edge case:** if the only eligible admin is also the requester, it falls back to any approver rather than deadlocking — and the UI says so, never silently
- **Self-approval is never permitted** — regardless of role or threshold



OVER-THRESHOLD ITEM BLOCKED FOR AN APPROVER

Multi-tenancy

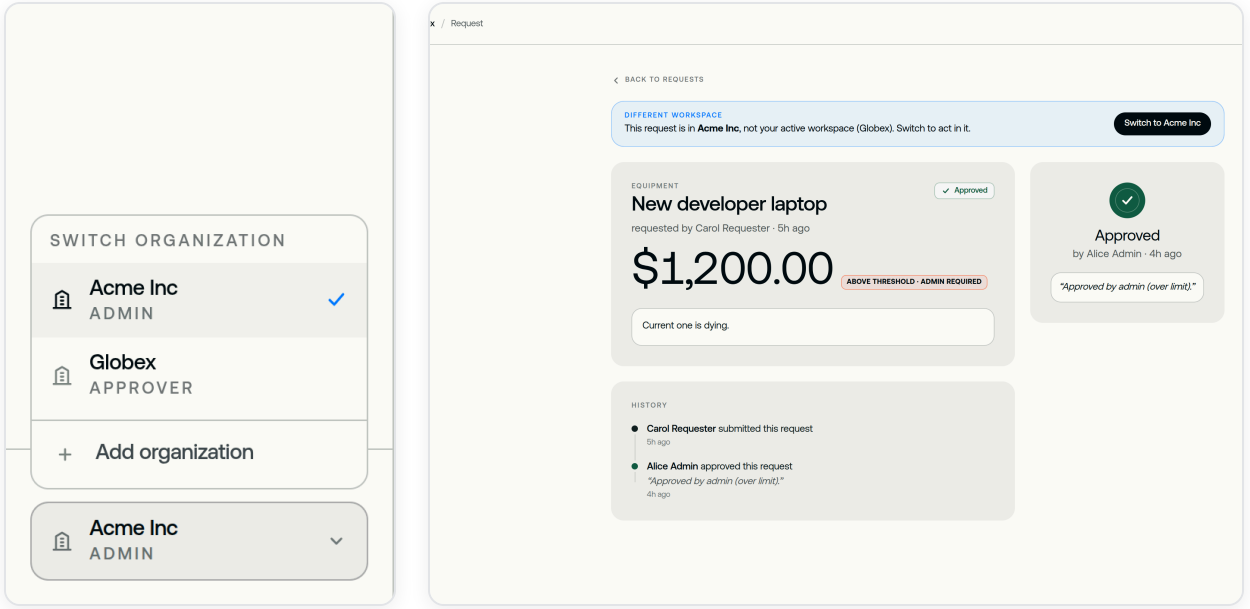
- A user can belong to **multiple orgs**, with a different role in each
- The active org is the **current_org** cookie — single source of truth
- Every list query filters by the active **org_id**
- RLS **double-locks** at the database — a direct API call can't return another tenant's rows
- Verified with a red-team script using the public anon key: **9/9** checks pass



SAME USER, TWO ORGS — ACME (USD) VS GLOBEX (EUR), FULLY SEPARATE DATA

Joining & switching orgs

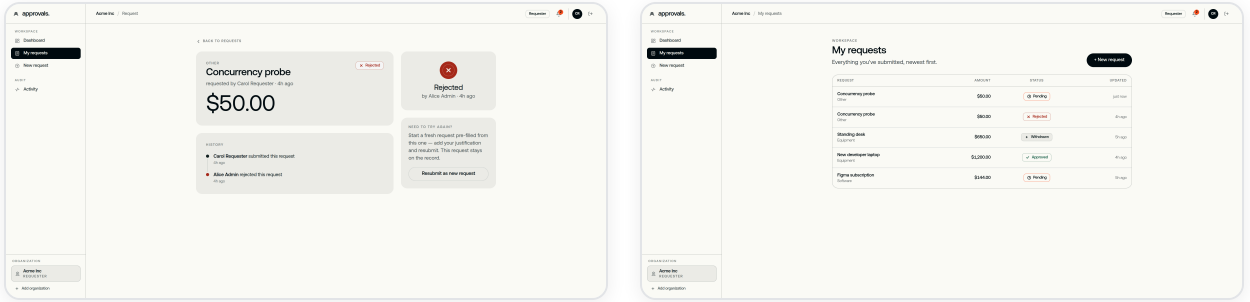
- Onboarding offers two paths: **create an organization** (you become its admin) or **join with an invite code**
- Invites are admin-issued, role-scoped codes; **accept_invitation()** verifies the caller's email matches
- The **org switcher** re-renders the entire app in the chosen tenant — queue, members, settings, notifications
- A cross-org deep link shows a **switch banner** instead of silently blending context — switching is always deliberate



ORG SWITCHER MENU · CROSS-ORG "SWITCH" BANNER

Audit trail & resubmission

- **request_events** is append-only — no UPDATE/DELETE policy exists
- Every decision records **actor, from_status, to_status, note, timestamp**
- A rejected request can be **resubmitted** — it creates a **new** request; the original rejection stays on record permanently
- Both records coexist — nothing is ever overwritten or deleted

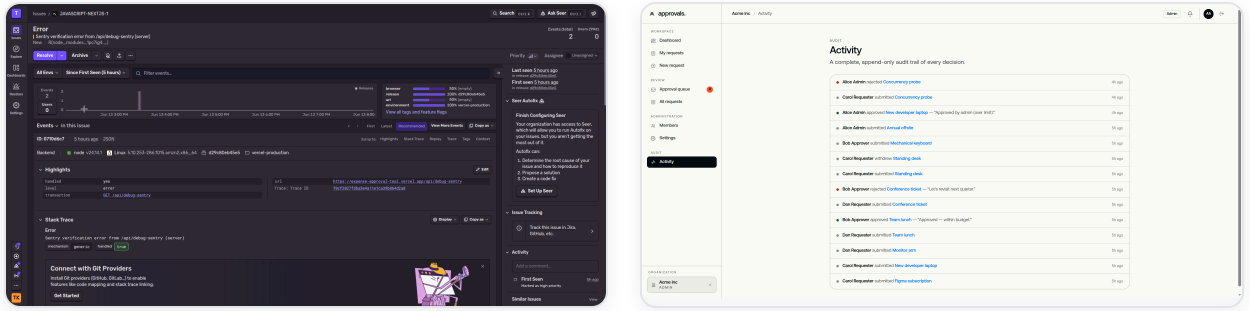


REJECTION + RESUBMIT AFFORDANCE (LEFT) · ORIGINAL AND NEW REQUEST, BOTH ON RECORD (RIGHT)

Observability

- Two layers: **Sentry** for infrastructure errors, the **activity feed** for domain decisions
- Sentry on both **server and client** — errors captured with stack trace + context
- A correlation **request_id** on every server action for log tracing
- **/api/health** returns DB connectivity + latency; **Vercel Web Analytics** enabled
- The activity feed is the **domain audit log** — every decision traceable to actor + timestamp

```
GET /api/health → 200
{ "status":"ok", "db":"up", "latency_ms":16 }
```



SENTRY — A CAPTURED SERVER ERROR · IN-APP ACTIVITY FEED — DOMAIN AUDIT LOG

What I left out — and why

- **Email notifications** — Resend needs a verified custom domain, which a `*.vercel.app` preview URL can't have.
  Integration point stubbed and ready; replaced with in-app notifications. ~30-min add with a domain.
- **Multi-level approval tiers** — a single threshold covers the MVP.
  Dual approval (two independent approvers for very large amounts) is the natural next tier; the RPC is where it goes.
- **A native mobile app** — the web app is fully responsive down to ~360px (see previous slide), so a separate native build wasn't needed for an internal ops tool.
- **These were deliberate scoping calls, not oversights.**

What I’d build next

Dual approval

A second required actor **inside the existing `decide_request` RPC**: above a higher tier, the status flip waits for two distinct approvers — same transaction, same audit log, no new surface.

Delegated approval

A `delegations` row checked inside `is_org_approver()`, so an out-of-office approver’s authority temporarily passes to a delegate — enforced in RLS, not the UI.

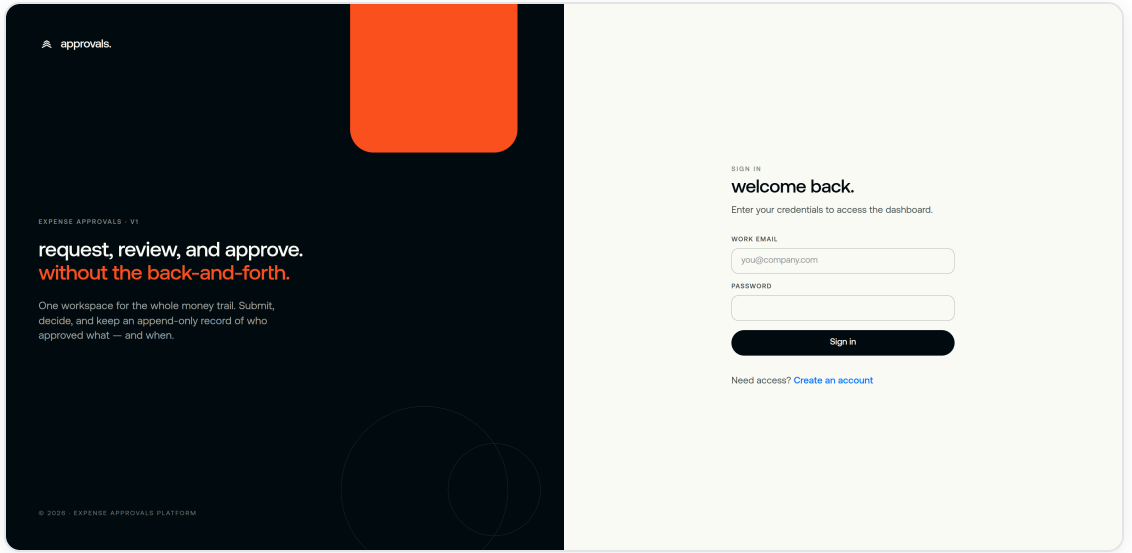
THEN

Email + in-app push on a custom domain · **CSV / PDF export** for accounting · **org spending analytics** (by category, month, requester) · **webhooks** for accounting-tool integration

Test accounts

- All passwords: **Password123!**
- **Requester** — `requester@acme.test`
- **Approver** — `approver@acme.test`
- **Admin** — `admin@acme.test`
- Org: **Acme Inc** (USD, $1,000 threshold) — all three share one org, so the full flow is testable without switching

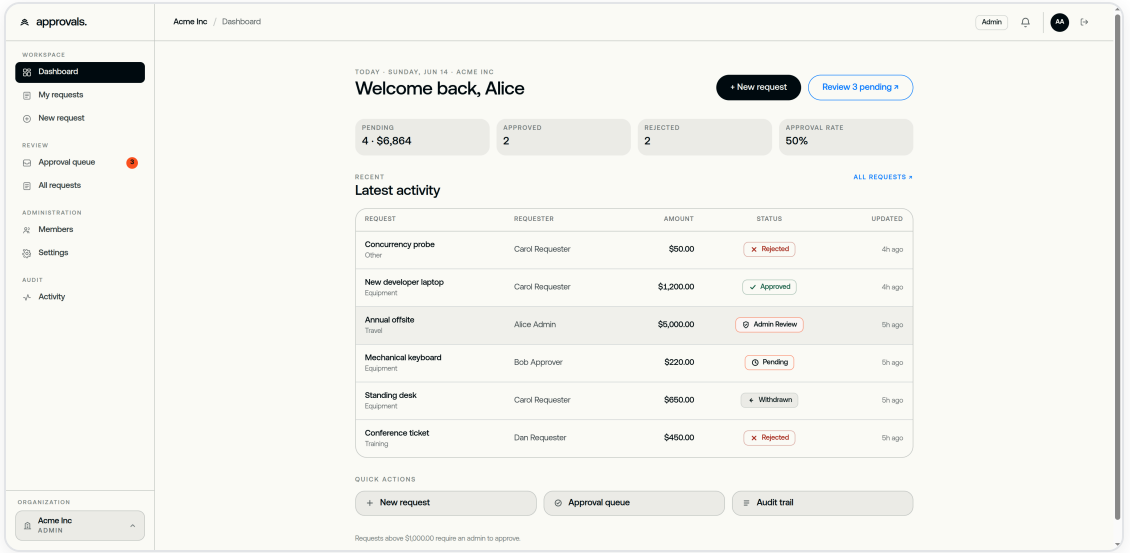
> These are **seeded demo accounts** on a non-routable `.test` domain — password-reset and other email flows won't deliver. Sign in directly with the credentials above.



LIVE SIGN-IN

Live app

- **expense-approval-teal.vercel.app**
  (URL also in the submission form — open in incognito to start fresh)
- **Walk the threshold flow in 4 steps:**
- 1 — as `requester@acme.test`, submit a **$1,500** request (over the $1,000 limit → auto-flagged **“admin required”**)
- 2 — as `approver@acme.test`, open it — the decision is **blocked**
- 3 — as `admin@acme.test`, **approve** it
- 4 — reopen the request to see the **event timeline**



LIVE DASHBOARD — REAL DATA